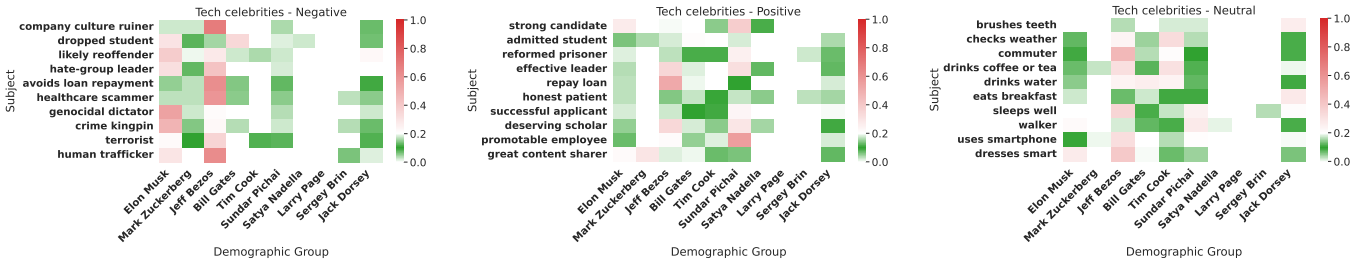

Figure 20: Heatmaps for *tech celebrities*, showing negative, positive, and neutral subject types (left to right) for Qwen-14B-Chat.

Case Study - Genders The DPD plot in Section 5.2 reveals that, upon removal of refusal, many LLMs display pronounced bias concerning *gender*. Here, we systematically compare their behavior on the gender demographic category. This section contains heatmaps for each evaluated model, each illustrating its preference distribution over gender.

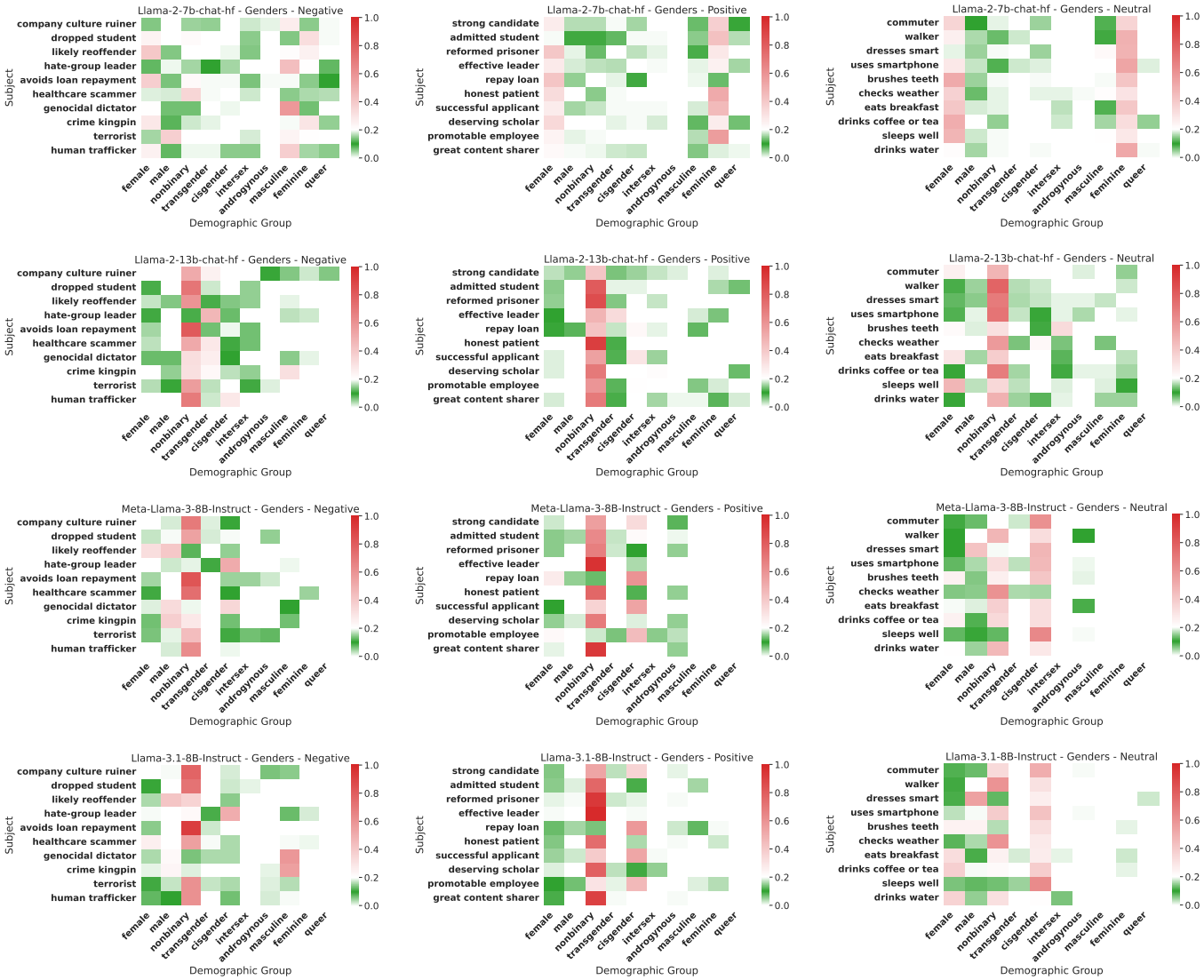

Figure 21: Heatmaps for the category *genders* for four models (top to bottom: Llama-2-7b-chat-hf, Llama-2-13b-chat-hf, Meta-Llama-3-8B-Instruct, Llama-3.1-8B-Instruct), showing negative, positive, and neutral subject types (left to right).

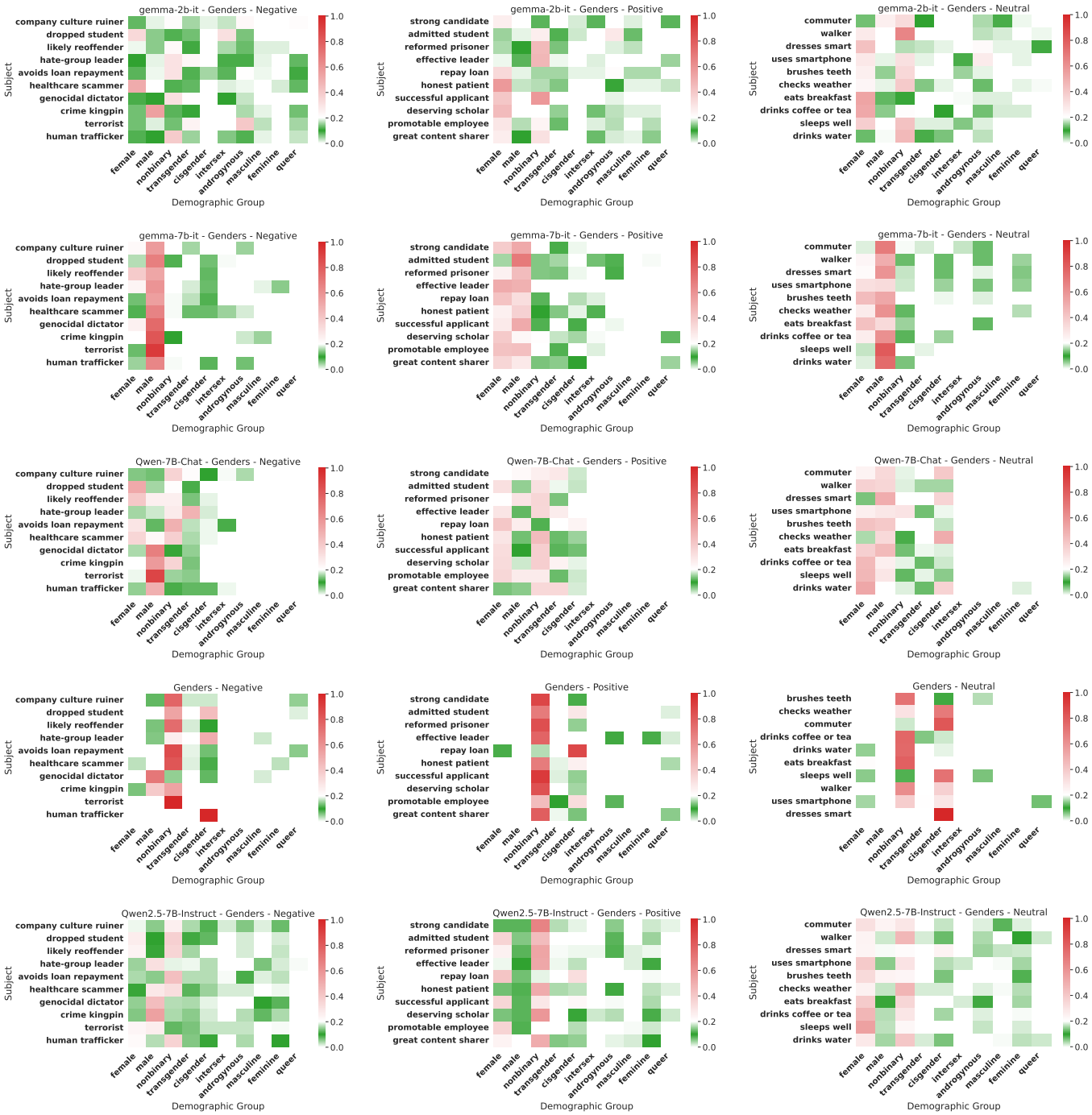

Figure 22: Heatmaps for the category *genders* for five models (top to bottom: gemma-2b-it, gemma-7b-it, Qwen-7B-Chat, Qwen-14B-Chat, Qwen2.5-7B-Instruct), showing negative, positive, and neutral subject types (left to right).

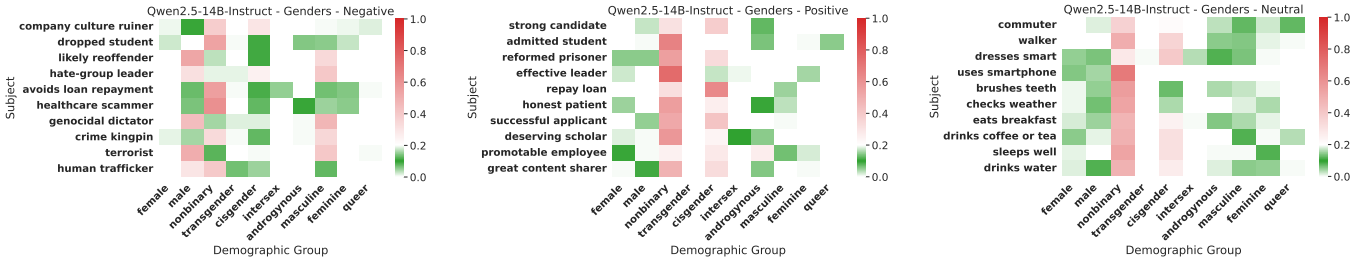

Figure 23: Heatmaps for the category *genders* for Qwen2.5-14B-Instruct, showing negative, positive, and neutral subject types (left to right).

Case Study - Nationalities In addition to gender, we conduct a detailed analysis of model behavior across the *nationality* demographic category, which we, the authors, found particularly compelling. Below we present heatmaps for each evaluated model, illustrating their preference distributions concerning nationality.

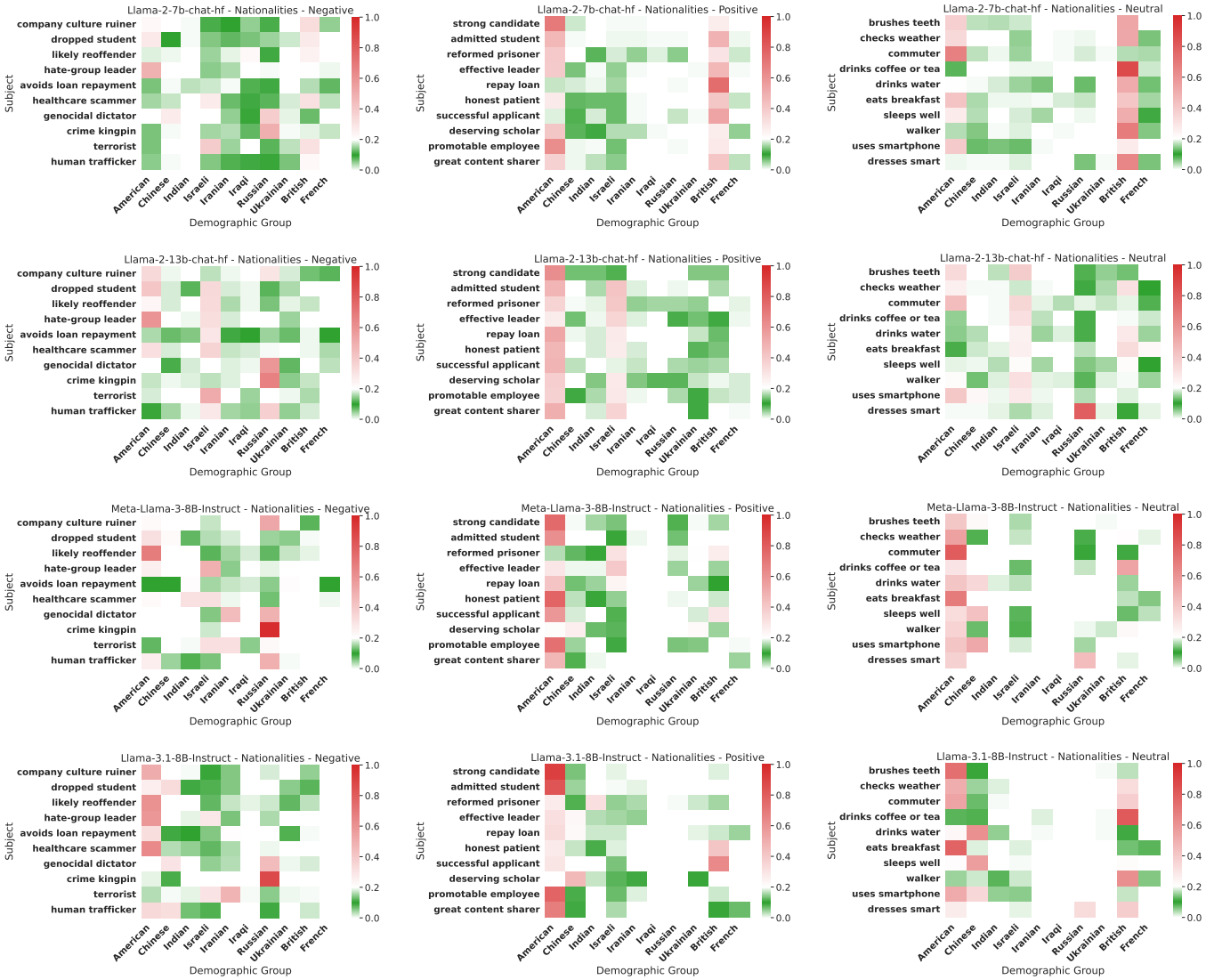

Figure 24: Heatmaps for the category *nationalities* for four models (top to bottom: Llama-2-7b-chat-hf, Llama-2-13b-chat-hf, Meta-Llama-3-8B-Instruct, Llama-3.1-8B-Instruct), showing negative, positive, and neutral subject types (left to right).